



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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ASSIGNMENT No: 6

TITLE: Write a Java program to demonstrate the use of an inner class, an anonymous class.

Problem Statement

Write a Java program to demonstrate the use of an inner class, an anonymous class.

Learning Objectives

To implement inner classes and anonymous classes for improving encapsulation and event handling in Java.

Theory

An inner class is a class defined inside another class and has access to all members of the enclosing class, including private members. Inner classes help improve code organization and logical grouping of related classes. Anonymous classes are unnamed inner classes that are instantiated only once and are mainly used for implementing interfaces or abstract classes. They simplify code by eliminating the need for separate class definitions. These concepts are widely used in Java GUI programming and event handling.

Outcome

Conclusion

In this experiment, we successfully implemented two Java programs using inner classes and anonymous classes. We learned that an inner class helps keep related code together and can access the outer class's data. We also learned that an anonymous class is useful for creating a one-time object without writing a separate class. This experiment improved our understanding of Java's object-oriented programming concepts and their practical use.



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Exercise

1. Create a Vehicle program where an inner class displays vehicle details and anonymous class performs an action.

Input

```
1 class Vehicle {
2     String brand = "Toyota";
3     String model = "Fortuner";
4     int year = 2024;
5     class VehicleDetails {
6         void display() {
7             System.out.println("Vehicle Details");
8             System.out.println("Brand : " + brand);
9             System.out.println("Model : " + model);
10            System.out.println("Year : " + year);
11        }
12    }
13 }
14 interface Action {
15     void drive();
16 }
17 public class VehicleDemo {
18     public static void main(String[] args) {
19         Vehicle vehicle = new Vehicle();
20         Vehicle.VehicleDetails details = vehicle.new VehicleDetails();
21         details.display();
22         Action a = new Action() {
23             public void drive() {
24                 System.out.println("The vehicle is now driving.");
25             }
26         };
27         a.drive();
28     }
29 }
30
```

VehicleDemo [Java Application] C:\Users\pdave\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.11.v20260515-1531\jre\bin\javaw.exe (01-Aug-2026, 12:58:33 pm)
Vehicle Details

Output

```
Vehicle Details
Brand : Toyota
Model : Fortuner
Year : 2024
The vehicle is now driving.
```



2. Develop a Food Delivery Application where an inner class handles order details and anonymous classes handle delivery status updates.

Input

```
FoodDelivery.java
1 class Order
2 {
3     private String foodName;
4     private int quantity;
5     Order(String foodName, int quantity)
6     {
7         this.foodName = foodName;
8         this.quantity = quantity;
9     }
10    class OrderDetails
11    {
12        public void displayOrder()
13        {
14            System.out.println("Food Item : " + foodName);
15            System.out.println("Quantity : " + quantity);
16        }
17    }
18 }
19 interface Delivery
20 {
21     void updateStatus();
22 }
23 public class FoodDelivery
24 {
25     public static void main(String[] args)
26     {
27         Order order = new Order("Pizza", 2);
28         Order.OrderDetails details = order.new OrderDetails();
29         System.out.println("==== ORDER DETAILS =====");
30         details.displayOrder();
31         System.out.println();
32         Delivery preparing = new Delivery()
33         {
34             public void updateStatus()
35             {
36                 System.out.println("Status : Order is being prepared.");
37             }
38         };
39         Delivery outForDelivery = new Delivery()
40         {
41             public void updateStatus()
42             {
43                 System.out.println("Status : Order is out for delivery.");
44             }
45         };
46         Delivery delivered = new Delivery()
47         {
48             public void updateStatus()
49             {
50                 System.out.println("Status : Order delivered successfully.");
51             }
52         };
53         preparing.updateStatus();
54         outForDelivery.updateStatus();
55         delivered.updateStatus();
56     }
57 }
```

```
32 Delivery preparing = new Delivery()
33 {
34     public void updateStatus()
35     {
36         System.out.println("Status : Order is being prepared.");
37     }
38 };
39 Delivery outForDelivery = new Delivery()
40 {
41     public void updateStatus()
42     {
43         System.out.println("Status : Order is out for delivery.");
44     }
45 };
46 Delivery delivered = new Delivery()
47 {
48     public void updateStatus()
49     {
50         System.out.println("Status : Order delivered successfully.");
51     }
52 };
53 preparing.updateStatus();
54 outForDelivery.updateStatus();
55 delivered.updateStatus();
56 }
57 }
```



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Output

```
===== ORDER DETAILS =====  
Food Item : Pizza  
Quantity  : 2  
  
Status : Order is being prepared.  
Status : Order is out for delivery.  
Status : Order delivered successfully.
```